Dimethyl Methylphosphonate Responds Opposite to Soman, Cyclosarin and VX in a Colorimetric Array of Authentic Chemical Warfare Agents

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Abstract

Chemical warfare agent detection is qualified on simulants, and we asked what measurement establishes that one reproduces its agent. We reanalyse the colorimetric tables of three published runs of one 29-component array, which measured sixteen authentic agents at four concentrations in two optical channels. Dimethyl methylphosphonate, the standard nerve agent simulant, gives a response pattern positively correlated with sarin, $+0.65$ and $+0.57$ in the two channels, and anti-correlated with soman, -0.49 and -0.48, with cyclosarin, -0.72 and -0.49, and with VX, -0.70 and -0.27; tabun changes sign between channels. The two channels are read from the same wells and are not independent replicates, the comparison comes from one run, and the tables carry no

repeats, so we report this as a measurement and not as a test. The array does discriminate on average: in all six run-channel cells every compound agrees with itself across concentrations more closely than with the mean of the other compounds, and all six clear $q = 0.05$ against a label-permutation null. Descriptor distance, the basis on which simulants are selected, does not resolve the difference. Twelve exact Mantel tests and six combinations of them leave nothing below $q = 0.05$, but the design resolves a rank correlation of 0.85 at 80% power at the corrected threshold and the largest observed is 0.57, so that is a limit on the design, not a demonstration of absence. Across the three runs, one designated simulant has been measured beside the agents.

Introduction

Nearly every chemical warfare agent detector in the literature is developed, optimised and qualified on compounds that are not chemical warfare agents.[1,2] Dimethyl methylphosphonate stands in for the G-series nerve agents,[3] dipropylene glycol monomethyl ether for training detectors,[4] and methyl salicylate for vapour penetration through protective ensembles.[5,6] The substitution is unavoidable, because the agents are lethal at concentrations where handling them requires a licensed facility, and it is load-bearing: a sensor reported as selective for sarin has, in most cases, never been exposed to sarin.

What licenses the substitution is a claim of similarity. The compound chosen is said to be agent-like, and similarity is read from molecular descriptors: vapour pressure, dipole moment, frontier orbital energies, substructure and shape. We test the first of those separately, on measured vapour pressures, in Section . Cheminformatic selection on exactly those grounds has been proposed as the systematic way to find a simulant,[7] reviews catalogue the compounds in use for environmental behaviour,[8] for emergency decontamination trials[9] and for toxic industrial chemicals,[10] and the US Army maintains a test operations procedure governing simulant selection for laboratory, chamber and field testing.[11] The claim that matters for a detector is different. It is that the substitute produces the response the agent would

produce in the same device. Those two claims are connected only if descriptor similarity transfers to response similarity.

That transfer has been tested once, for a different measurement and in simulation. Agrawal and co-workers asked how effectively four common simulants, dimethyl methylphosphonate and diisopropyl fluorophosphate among them, mimic the adsorption properties of sarin and soman, by computing dilute-limit adsorption for all six compounds in 2969 metal–organic frameworks of known structure under two independent force fields.[12] That design answers the question for adsorption, which is the quantity a capture material is judged on. It cannot answer it for detection, because the response of a detector is not an adsorption isotherm, and because a simulation of the agent is not a measurement of it. What has not been done is the measured comparison: the agents themselves, in a detector, against the descriptor distances on which the substitution rests.

It has not been measured because measuring it requires the agents. Response data on authentic agents exist in few places, and the compounds used as simulants are rarely run alongside them. A recent survey of threat simulants records that they have generally been selected on the basis of their historical use, and reports a draft ASTM guide for explosive and contraband simulants; it describes no equivalent document on the chemical warfare agent side.[13] A critical review of the low-toxicity simulants in current use states the selection criterion plainly, as the alignment of physicochemical properties such as hydrophobicity, volatility and solubility with those of the agent, and assesses dipropylene glycol monomethyl ether and malathion among others on that basis.[14] That criterion is a statement about the compound. It is not a statement about the detector, and nothing in the review measures whether a compound selected that way produces the response the agent produces.

Asking whether the components a detection claim rests on have themselves been validated is a question this field has begun to put in print: a recent guide to choosing and validating aptamers for biosensors opens on the observation that many reported aptamers remain insufficiently characterised, and that the resulting reproducibility problem makes a

reliable choice hard.[15] Simulants are the analyte-side form of that problem, and the concern itself is not new. It has been stated in this journal. Reporting a bioelectronic nose tested on authentic sarin and VX, two of us and our co-workers noted that detection research built on structural simulants may not carry over to the agents, depending on the selectivity of the sensor, and used combinatorial pattern recognition across a multi-receptor array to separate the agents themselves.[16] That work asserted the risk and worked around it by measuring the agents. It did not quantify the risk, because quantifying it requires putting the descriptor distances and the measured responses side by side and asking whether one predicts the other. The same holds for our group's field-deployable detection work, which likewise measured the agents rather than testing the substitution.[17] The contribution here is that measurement and not the assertion. The three source studies were run at an OPCW designated laboratory, which identifies and assays authentic agents under an accredited procedure. That is what made the original measurements possible; the reanalysis reported here handled no agent and made no measurement of its own.

This work reports that measurement. A 29-dye colorimetric array was used to record the responses of sixteen authentic agents, twelve of them Chemical Weapons Convention Schedule 1 compounds and four Schedule 3 industrial chemicals, at four solution concentrations in two optical channels, and those responses were compared against the molecular-descriptor distances that simulant selection relies on. The array separates compounds in every run and channel once the compound-axis mean is removed, which is the representation every comparison here uses; on the raw vectors, before that removal, the separation is not resolvable, and the cross-validated accuracies quoted for it come from runs that contain no simulant (Section). Descriptor distance nonetheless carries no detectable information about response distance. Dimethyl methylphosphonate, the compound the field treats as the G-agent surrogate, is anti-correlated with two of the four G agents, and its resemblance to sarin weakens as the solution concentration falls. Where the designation record and the regulatory toxicity axis might have supplied the missing evidence, the record names a measurement that is not

detection and the axis the agents are placed on does not reach the incumbents.

This is not a new sensing method, and we propose no replacement compound. It is a measurement of whether the substitutes the field already relies on carry the property they are relied upon for.

Results and Discussion

The array separates the agents

A comparison between descriptor space and response space is worth making only if the response measurement resolves the compounds at all. That question is already answered in the reports these data come from, and the answer does not depend on anything we do here. On the first run a linear discriminant classifier built on a six-dye subset under dual illumination separated the four G agents, VX and dimethyl methylphosphonate at 100% accuracy, against 87.5% under visible illumination alone.[18] On the second, a six-dye subset classified all seven agents at an accuracy of 0.964 with a macro F_1 of 0.964 under five-fold stratified cross-validation.[19] On the third, a four-dye subset separated the four Novichok agents at an accuracy of 1.000 and a macro F_1 of 1.000 under leave-one-out cross-validation, over a working range of 10 to 500 μM with a detection limit near 10 μM.[20]

Three qualifications move where the premise rests. Each figure is obtained on a dye subset chosen for that run, not on the 29-component array we analyse; the first run reports no held-out evaluation, so its 100% is an apparent rather than a validated accuracy; and that first run is the only one in which a designated simulant was measured beside the agents, so the two cross-validated figures come from runs that contain no simulant. We cannot reproduce either cross-validated figure from the deposited tables, which are averaged over replicates, so we quote them and do not verify them. The premise for the first run therefore rests on our own check below, which covers all six run-channel cells.

What our own check adds is narrower and is reported because the comparison that follows

is made on patterns rather than on class labels. Each exposure gives a 29-dye ΔE_{00} vector, which we scaled to unit length, centred dye-wise at each concentration, and scaled to unit length again, so that agreement is measured between patterns and not between magnitudes, and took agreement as the cosine. The question it answers is whether a compound's pattern is stable enough across dose to be treated as that compound's pattern at all.

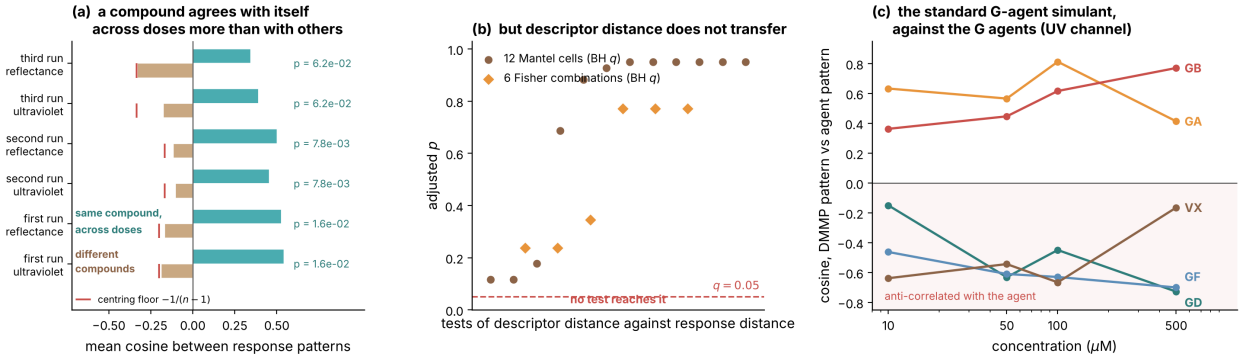


Figure 1: Response-space tests on authentic agents. (a) Mean cosine between response patterns, within one compound across concentrations and between different compounds, for three runs in two optical channels. (b) Adjusted p for every test of descriptor distance against response distance: twelve exact Mantel cells and six Fisher combinations of the independent runs. (c) Cosine between the response pattern of DMMP and those of the four G agents and VX, by concentration, in the ultraviolet channel.

It is stable enough (Figure 1a). One compound's patterns at two concentrations correlate at +0.345 to +0.542, different compounds anti-correlate at −0.103 to −0.330, and the ordering holds in all six run-channel cells. Because the independent unit is the compound and not the pair, we pre-specified a sign-flip enumeration over the four to seven compounds of each cell. That test returns the smallest value its compound count allows in all six cells, from 0.0078 to 0.0625, so what it reports is a property of the design and not of the data: centring leaves the per-compound difference positive for almost any input that is not degenerate. We withdraw it. The same statistic is tested here against a null that permutes compound labels within each concentration, which leaves every centred vector untouched and changes only which response is assigned to which compound. All six cells clear $q = 0.05$. Four of the six are sampled at 10,000 relabellings and return the smallest value that sampler can

give, 9.999×10^{-5}, so those four are censored at the sampler's resolution and the smallest adjusted value over the six, 0.00015, is an upper bound rather than an estimate. The two four-compound cells are exact, by enumeration of all 13,824 relabellings, and give one-sided values of 0.0046 and 0.0033. The effect is large enough that the censoring does not bear on the conclusion: the observed statistics run from +0.558 to +0.729 against null means of +0.102 to +0.314. Cohen's d_z over compounds runs from +0.92 to +5.12 as a descriptive effect size. Supporting Information Section S13 gives the per-cell values and why the bound the earlier version used does not apply.

The tables analysed here hold one value per compound, concentration, dye and channel, so the within-compound term is agreement of a compound with itself at a different dose and not agreement between repeated measurements. Repeats were reported in the source work for the second and third runs, for selected dye-analyte combinations in the second[19] and for all experiments in the third,[20] but the repeat-level values are not in these tables, so no repeatability estimate enters any analysis below.

Two features of the measurement belong here rather than in the Supporting Information. The first is that dye-wise centring drives the mean between-compound cosine negative by itself, so an earlier version's comparison against $-1/(n-1)$ does not hold; Supporting Information Section S13 gives the two reasons. What does quantify the inflation is the permutation null. Relabelling compounds within each concentration leaves a mean within-minus-between difference of +0.1024 to +0.3138 across the six cells, against observed differences of +0.5578 to +0.7288. What the test establishes is the per-compound statement, not the stronger pairwise one: in 13 of the 34 compound-and-cell instances, some other compound's consensus pattern is closer to a compound than that compound's own agreement across concentrations. The second is that six dye columns in the ultraviolet pivot of the second run were bit-identical across every row, because those dyes do not fluoresce under ultraviolet excitation and the regions of interest had read plate background. Columns identical across all rows are counted once, which leaves 23 effective channels in that cell and 25 in the ultraviolet pivot of the

third run. Supporting Information Section S22 sets out what the array does and does not establish in more detail.

The standard simulant responds opposite to three of the five agents measured with it

The array resolves the compounds, so the measurement can be put to the one compound the field uses in their place. DMMP is named here rather than treated as one of many, because the field treats it as the G-agent surrogate and because decontamination efficiency is quantified against it. The first run contains DMMP together with the four G agents and VX, which is the only place in the three runs where a designated simulant was measured beside the agents it stands in for, so its response can be set against theirs directly.

In the ultraviolet channel DMMP correlates with tabun at +0.764 and with sarin at +0.645, and anti-correlates with soman at −0.492, cyclosarin at −0.721 and VX at −0.702. In the reflectance channel the same comparison gives −0.112, +0.569, −0.479, −0.495 and −0.273. The mean over all four G agents is +0.049 in the ultraviolet channel and −0.129 in reflectance; over the three G agents other than tabun it is −0.189 and −0.135. Neither mean carries the claim on its own, because the sign of the correlation changes with the agent and with the channel: DMMP correlates with tabun and sarin in the ultraviolet and anti-correlates with soman and cyclosarin, and tabun changes sign between the channels while sarin does not. What this measurement does not show is a compound that behaves like the G series as a class.

A weaker claim survives in one channel. If DMMP is G-like at all it should sit closer to the G series than to VX, and that contrast, taken against the mean over the three matched G agents and excluding tabun, is +0.513 in the ultraviolet channel and +0.137 in the reflectance channel. The ranges quoted with them, +0.131 to +0.722 and −0.144 to +0.290, come from resampling the four design concentrations with replacement. Four fixed design levels are not an exchangeable sample, so these are descriptive ranges and not tests: the ultraviolet range

lies above zero and the reflectance range spans it.

Concentration changes the picture (Figure 1c). The DMMP-to-sarin cosine rises monotonically from +0.363 at 10 μM to +0.771 at 500 μM, and the DMMP-to-cyclosarin cosine falls monotonically from −0.462 to −0.700 over the same range. Monotonicity is perfect in three of the ten agent-channel series. It is also the strongest result the design can give: with four concentrations the two-sided exact permutation has a floor of $p = 0.083$, three series reach exactly that floor, and Benjamini–Hochberg correction over the ten leaves a smallest adjusted value of 0.278. We report the direction and not a significance claim.

Two limits sit on that direction. The concentrations are of analyte in a dye solution, not of vapour at an inlet, so we do not translate the range into the operating band of a point detector. And the mean response magnitude rises with concentration at a Spearman correlation of exactly +1.000 in every one of the six run-channel cells, so a similarity that grows with concentration cannot be separated here from one that grows with signal-to-noise ratio; distinguishing them needs a repeat-level noise estimate the deposited tables do not carry. What the direction supports is that the resemblance between this simulant and this agent is not constant over the range at which the array was characterised.

Two further readings suggested themselves and did not survive their own controls. We report both, because the alternative is to present artefacts as results. The first was a ranking. DMMP is the nearest of the five candidates to tabun in the ultraviolet channel and to sarin in the reflectance channel and to nothing else, which reads as evidence that the optical channel decides which agent the standard simulant stands in for. It is not. With five candidates the expected number of first places over ten agent-channel cells is 2.0, the observed number is two, and the one-sided binomial p is 0.62. The ten cells share DMMP's two response vectors, so they are not ten independent trials and the binomial is an approximation; the observed count sits on the expectation, where no correction could move the reading.

The second was a channel decomposition. The cosine is a sum over dye channels, and the DMMP-to-tabun value of +0.764 turns out to rest on two of the 29 channels, with the largest

single channel carrying 44% of the positive mass. Read alone, that makes the resemblance look fragile. Two controls show it is not a property of DMMP. Agent-to-agent pairs in the same run concentrate identically, with median two channels in the ultraviolet and one in the reflectance channel, and a null built by permuting the dye labels of the DMMP vector gives a mean of 2.00 and 2.26. Concentration of this kind is what the product of two centred unit vectors in 29 dimensions does, and it says nothing about this compound.

What remains after both controls is the set of measured values themselves, and one structural fact. The first run contains no candidate simulant other than DMMP, so the designation cannot be contrasted against an alternative in response space at all. The permutation null that the descriptor analysis of the next section relies on is, in response space, empty.

Descriptor distance does not resolve the difference

One compound is one compound. The practice that put DMMP there is a general one, and it rests on an implicit transfer: that a compound close to an agent in descriptor space will be close to it in whatever the detector measures. We tested that transfer directly, as the agreement between two distance matrices over the same compounds, one built from descriptors and one from response patterns. Agreement was measured by Mantel correlation and referred to a null built by enumerating every relabelling of the compounds, so the test is exact rather than sampled.

The test is run within each run and never across runs. The three runs share no compound, so the dye-wise sensitivity of one plate cannot be calibrated against another, and pooling responses across them would require a commutability assumption we cannot support. That restriction sets the power. Six compounds in the first run and six in the second carry both a response pattern and a descriptor vector, giving fifteen pairs in each, and the third run drops out because the A-series compounds are not in the descriptor panel, which follows from the source itself: it reports the four as the codes A-230, A-232, A-234 and A-242, gives their

structures as representative rather than assigned, and does not state which compound each code denotes.[20] Nitrogen mustard drops out of the second run for the same reason.

Across the twelve cells, three spaces by two channels by two runs, nothing survives correction (Figure 1b). Three cells reach a nominal p below 0.05, all of them in the second run: the structural space in the ultraviolet channel at 0.0167, the merged space in the same channel at 0.0194, and the electronic space in the reflectance channel at 0.0444. Benjamini–Hochberg correction over the twelve leaves a smallest adjusted value of 0.1167. The pattern of the three is itself uninformative about transfer, because the two ultraviolet cells implicate the structural and merged spaces while the reflectance cell implicates the electronic space, and the same space does not reproduce across channels of the same plate.

There is a further asymmetry in where those three cells sit, and it matters more than their size. The second run contains seven authentic agents and no designated simulant. Whatever agreement those three cells carry is agreement between one agent and another, measured on one plate, and it is not evidence that a simulant reproduces an agent. The only run that contains a designated simulant beside the agents is the first, and the first run gives nothing in any space or channel, with $|\rho| \le 0.225$ and every p at or above 0.229. The one run that could speak to simulant fidelity is silent, and the run that speaks is not about simulants.

Combining runs does not change the answer. The first and second runs share neither compounds nor plates, so their exact Mantel p values are independent and can be combined by Fisher's method. We report all six combinations of three spaces and two channels rather than selecting one, which would reintroduce the analyst freedom the test is meant to remove. The smallest adjusted value over the six is 0.2373, and none reaches 0.05. Section S14 of the Supporting Information gives the full permutation family, and Section S11 the six multiplicity schemes under which it was read.

What this does not license is the claim that no transfer exists. Fifteen pairs is a weak test, and the pooled rank correlations are correspondingly imprecise: the structural space in the ultraviolet channel gives $\rho = +0.31$ with a 95% interval from -0.45 to $+0.81$, which

excludes neither zero nor a strong association. The defensible statement is that the transfer on which simulant selection depends has not been established, in the one body of authentic-agent response data where it could be tested, and that the experiment which would settle it is a run containing a compound already measured in an existing run. Such a bridge would fix the plate correction and open the 136 cross-run pairs that are closed today.

Descriptor space cannot supply the missing validation

If response space cannot establish the transfer, the question becomes whether descriptor space can at least establish the premise it is used for, that the incumbents sit closer to the agents than other compounds do. This can be asked exactly. Designating three compounds out of the non-agents in a panel is a choice among a finite set, so the incumbent designation can be referred to every alternative designation rather than to a sampled null. The primary panel of thirty-five compounds gives 2024 designations and the reduced panel of twenty-eight gives 680.

On the primary panel the answer is flat (Figure 2). No agent of the eleven has a valid match, the smallest nominal p over the eighteen tests is 0.1917, and the Benjamini–Hochberg adjusted value is 0.6299.

What produces that zero is worth stating, because it is not chemical background. The criterion has two parts, a normalised distance of 2 or less and a distance smaller than that to every compound outside the designated set, and on this panel it is the second part that fails. Sarin, soman and cyclosarin all sit inside the threshold of DMMP, at 1.1805, 1.4583 and 1.6041. Each of them fails because diisopropyl fluorophosphate is nearer still, and tabun, which is outside the threshold at 2.5320, has diethyl cyanophosphonate nearer than DMMP. Those two compounds are not interferents. They are compounds the sensing literature itself proposes as simulants. If the comparison set is narrowed to the sixteen interferents, which is how the second part is implemented in the deposited distance table, the same three G agents pass and the count on this panel is three rather than zero. We report the wider reading as

the primary one because it is the criterion stated in Section , and we report both counts because the difference between them is not a rounding matter. The designated simulants are not failing to stand out from chemical background on this panel. They are being displaced by alternatives the literature has already put forward.

Nothing in that panel supports the designation of the incumbents in particular, and the reason it does not is that better candidates exist in descriptor space.

The reduced panel, which omits seven compounds, five of them named as simulants in the literature and two of them pesticides, reads differently. Three agents acquire a valid match, and the merged space returns a nominal one-sided p of 0.0250 with the structural space at 0.0279. Whether that survives depends on which correction is applied. Benjamini–Hochberg over the eighteen tests gives 0.1632, correcting within each space gives 0.0772, and the Westfall–Young single-step procedure, which uses the shared permutations and so handles the dependence between tests exactly, gives 0.0794. Only the scheme that fixes one statistic in advance returns a value below 0.05, at 0.0419. Significance here is a property of one correction scheme on one panel construction, and we report all four rather than the one that passes.

The surviving structural value is also an ordinary draw from the choices available to an analyst: drawing three descriptors at random from the structural block 2000 times places the observed value at the 22.4th percentile of that distribution. One quantity in this family does separate the incumbents, and it is easy to over-read. Across 500 random three-descriptor subspaces the incumbent designation reaches $p < 0.05$ in 27.4% of them, which ranks seventh of 680 ($p = 0.0103$) against a mean of 4.5%. That is a statement about the flexibility of the comparison, not about proximity to the agents. The nulls, the dimensionality controls and the block-weighting check are in Supporting Information Sections S16, S17 and S18.

One analyst freedom we did not enumerate when we set the comparison up moved the answer further than any we did. The electronic descriptors throughout are GFN2-xTB. Recomputing all three coordinates of the electronic space at B3LYP/def2-SVPD moves it

from one valid match of eleven and a one-sided p of 0.275 to five matches and 0.0368. Only one of the five is a pairing the designation record makes. We do not promote that value: it is a single uncorrected statistic in one space, reached by changing a method after seeing the first answer, on different geometries. What it establishes is that the space in which the incumbents did worst is the space whose answer the choice of electronic-structure method decides. The merged space does not move at all: thirty-eight coordinates, three valid matches and a one-sided p of 0.0250 under both methods. Supporting Information Section S10 gives the full comparison and the limits on it.

Panel composition decides the answer as well. Adding back the five compounds the sensing literature also treats as simulants moves the count from three valid matches to zero, because diisopropyl fluorophosphate is nearer to three G agents than any incumbent is, and 2-chloroethyl ethyl sulfide is nearer to sulfur mustard than dichloromethane is. The incumbents do not move. What moves is whether anything closer is allowed into the comparison. Resampling the descriptor columns a thousand times puts the same point in distributional form (Figure 2b): the three incumbents return a median of zero valid matches, with no match at all in 61.2% of replicates and at most one in 85.6%, while the eight-compound set returns a median of five; the difference is positive in every replicate.

So does the way the descriptors are encoded, and that freedom is the largest of the four. Four of the thirty-eight retained coordinates are binary indicators carried by exactly one compound of the twenty-eight: an arsenic count for lewisite, a phosphorus–cyanide count for tabun, a phosphorus–sulfur count for VX and a nitro count for chloropicrin. A binary column with a single one has sample variance exactly $1/n$, so standardising it puts the carrier and every non-carrier $\sqrt{n}$ apart and adds exactly n, here 28, to the squared distance between that one compound and every other. The addition says that the compound is unique in the panel, not that it is far from DMMP, and for tabun it is 47.6% of the squared distance to DMMP. Dropping those four coordinates moves tabun from 2.0038 to 1.4509 and the count from three valid matches to four. Dropping every binary indicator carried by three

compounds or fewer gives five, and dropping every coordinate with five or fewer distinct values gives six, with the merged-space one-sided p falling from 0.0250 to 0.0074. We kept the encoding that produces the smallest count, and we report the sweep because that is the only evidence that we did not select the encoding for its answer. Supporting Information Section S7 gives the column-by-column accounting.

A fifth choice is the shape block. Nine of the retained coordinates on each panel are functions of the three principal moments of inertia, and the correlation filter does not remove them because the relations are ratios rather than linear. We kept the full block and recomputed everything with the seven derived coordinates dropped, as a declared sensitivity rather than a replacement: the valid-match count on the twenty-eight-compound panel falls from three to zero, and on the primary panel it is zero either way. Supporting Information Section S5 gives the correlations, the contribution shares and the Mantel values under the reduced block.

Five analyst choices therefore move the count of valid matches: the panel, the order in which the correlation filter is handed its columns, the conformer embedding seed, the descriptor encoding and the shape block. None of them moves the result of Section . Recomputing the twelve Mantel tests under the encodings above leaves the smallest Benjamini–Hochberg value at 0.1167, 0.4722 and 0.3889, and dropping the derived shape coordinates raises it to 0.1708 with the largest rank correlation unchanged at 0.571. None is below 0.05, and removing the single-carrier indicators makes the agreement weaker rather than stronger. We draw the distinction explicitly because the two results are not equally load-bearing. The agreement test between descriptor distance and measured response survives every choice we enumerated. The count of valid matches survives none of them, and a quantity that five arbitrary choices can move between zero and six is not a measurement of anything. That is this paper's argument applied to its own instrument, and it is the reason we do not report a match count as a finding.

One descriptor the opening of this paper names first is absent from the panel above, and

we tested it separately rather than argue around it. Volatility enters as one axis, $\log_{10}$ of measured vapour pressure. Two tests were fixed before the values were seen and are held outside the family of seventy-two. On the pairing question the incumbent trio does beat chance, with a one-sided p of 0.0206 over all 680 designations, but the margin is 0.0015 in standardised units and the count of valid matches returns one match at $p = 0.4647$, so the axis does not qualify the incumbents either. On agreement with measured response it is the one quantity in this paper that reaches $q = 0.0430$ in the reflectance channel. That value sits in a two-channel family we adopted after withdrawing the prespecified statistic for this test, a single Fisher combination over the four evaluable cells, which returned $p = 0.0809$; we withdrew it because the two channels of one plate are not independent, not because of its value, but the family that replaces it is post-hoc and we say so. The same four numbers read as one family give 0.1389, and corrected jointly with the twelve descriptor cells, over sixteen, 0.1556. Nothing clears 0.05 under either of those two readings. Supporting Information Section S15 sets the axis out in full, with the distances, the four evaluable cells and why we do not read any of them as a departure from a designated pairing.

Neither the designation record nor a toxicity model fills the gap

Two places remain where the missing evidence might be found, and neither supplies it for the incumbents. We looked because the published criterion for choosing a low-toxicity simulant is a physicochemical one,[14] and a physicochemical criterion does not by itself say what measurement the compound was validated against or how toxic it is.

The first is the standard-setting record. We traced each incumbent to the documents that designate it. DMMP is identified in a NIOSH programme for respirator materials resistance testing, where it is named a filtration simulant for GB and for HD and the properties that carry the choice are mass transfer and absorbent capacity.[3] The same document names carbon tetrachloride and cyclohexane for the same purpose, so DMMP is one of three rather than the designated compound. It carries a draft-for-discussion marking, which is worth

saying plainly: the most-cited designation of the compound this field treats as the standard nerve agent simulant has never been issued as a final standard. That document also gives its reasons, and they are worth quoting because they are not what a detection user would assume. DMMP is called a good filtration simulant because it is relatively non-toxic, is easily detectable, and has a volatility intermediate between GB and HD. The third reason is the one that matters here: the volatility was chosen to sit between two agents rather than to match either, so a gap between DMMP and any single agent on that axis is the designation behaving as specified and not a defect in it. The second reason shows that detectability did enter the choice, as a property of the compound in an unspecified instrument, not as a measured correspondence between the compound's response and an agent's. It is designated a second time, for a different agent and a different measurement, in a National Research Council assessment of the Man-in-Simulant Test programme, which lists it as the fabric-testing simulant for VX and gives di-isopropyl methyl phosphonate for GB.[5] That report also states the basis on which its simulants were chosen, low toxicity and similarity of certain physical properties, and records that the chemical structure may be quite different. One compound therefore stands in for VX in one document and for GB and HD in another, against two measurements, neither of which is detection. DPGME appears in a US Army patent for a training composition, where the property required of it is detectability by ion mobility spectrometry.[4] For dichloromethane we found no designating document in the public record we searched; Supporting Information Section S8 gives the sources, the date and the three limits of that search, one of which is a Test Operations Procedure we identified but could not obtain. Each of the two traceable designations is a validation against a measurement. For DMMP those measurements are filter filtration and fabric penetration, neither of which is detection. For DPGME the measurement is detection, but by ion mobility spectrometry, a different transduction from the colorimetric response reanalysed here. A simulant is validated for a purpose, and that validation does not travel with the compound to a different purpose.

The second is regulatory toxicity (Figure 3). The axis on which the agents are placed

here is the EPA acute exposure guideline level, and nineteen of the twenty-eight panel compounds carry one. Two of the three incumbents do not. DMMP, the compound against which decontamination efficiency is quantified, has no AEGL, and neither has DPGME; dichloromethane has one. Those two are not absent from every register. The US Department of Energy protective action criteria give DMMP 11, 120 and 730 mg m^{-3} for PAC-1 to PAC-3, derived as temporary emergency exposure limits from a rat one-hour LC_{50}, and NIOSH gives DPGME an immediately dangerous to life or health value of 600 ppm with a recommended exposure limit of 100 ppm. What is missing is not a number of any kind but a number on the axis the agents occupy: those values rest on different averaging times, different severity definitions and different derivations, so placing them beside a ten-minute AEGL-3 would compare quantities that are not the same quantity. Supporting Information Section S24 lists what each register holds for the three incumbents with its source. A model cannot close the gap on the AEGL axis either. A random forest trained on 230 AEGL chemicals reaches $R^2 = 0.458$ in cross-validation, with an out-of-fold bias of +0.001 log units, so nothing in its internal statistics warns a user off. Applied to twenty military agents held out of training it is wrong in one direction without exception, by 0.40 to 3.42 log units. Its applicability-domain check certifies fourteen of the twenty as inside the domain, and the worst of those fourteen is wrong by 2.68 log units, a factor of 482. The mean error over the twenty is +1.68 log units with Cohen's $d_z = +1.96$. The measured axis and its gaps are tabulated in Supporting Information Section S25, and the test of whether fidelity and acute toxicity are associated at all, which finds no monotonic relation across the compounds carrying both axes, is in Section S26. Section S28 reads what that implies for the incumbent's position on the two axes, and states why we do not turn it into a recommendation.

Conclusions

The compound this field uses to stand in for the G agents responds opposite to three of the five agents it was measured beside, in both optical channels. That is one run and one array, and we report it as a measurement rather than as a test, but it is the measurement the practice has been missing. Simulant validity is not a property of a compound. It is a claim about a measurement, and the practice of designating chemical warfare agent simulants records neither the measurement a compound was validated against nor the comparison in which its validity was tested. The criterion the literature does state is physicochemical alignment with the agent,[14] which is a property of the compound and not of the measurement.

Three counts fix where the evidence stands. In the three runs reanalysed here, one designated simulant was measured beside the agents, in one run of one array, and we found no second co-measurement in the literature we searched; we state that as the limit of our search rather than as a claim about every measurement ever made. Twelve exact Mantel tests and six combinations of them leave no agreement between descriptor distance and response distance below $q = 0.05$, against a design that resolves a rank correlation of 0.85 at 80% power at that corrected threshold, so that count bounds the design rather than the chemistry; Supporting Information Section S13 gives the power simulation behind 0.85, and Section S20 the separate displacement analysis for the designation tests. On the primary panel no agent of the eleven has a valid match to an incumbent simulant, at a Benjamini–Hochberg adjusted value of 0.6299, and the compounds that displace the incumbents there are themselves literature-proposed simulants rather than interferents.

The documentary record accounts for the first of those counts rather than contradicting it. DMMP was designated against respirator filtration and fabric penetration, neither of which is detection, and DPGME against detectability by ion mobility spectrometry, which is a different transduction from the colorimetric response reanalysed here. A validation does not travel with a compound to a purpose it was never tested for. The toxicity axis stops in the same place: DMMP carries no AEGL, and the values other registers hold for it are not

the quantity the agents are placed on. A random forest trained on 230 AEGL chemicals is wrong in one direction without exception on twenty held-out military agents, so a prediction cannot stand in either.

Absence of evidence is not evidence of unsuitability, and we conclude nothing about whether the incumbents are poor choices. We propose no replacement compound either, because selecting one on descriptor similarity would repeat the error this paper is about.

Two readings this paper does not support should be ruled out, because the designating documents are cited here for what they measured. DMMP is the benchmark compound against which decontamination efficiency is quantified, and nothing in this work bears on whether it is suitable for that. Filter filtration and surface decontamination are mass-transport and reactivity measurements, our comparison is between a colorimetric response and a descriptor vector, and a compound can be a sound surrogate for one and an unsound surrogate for the other. The same holds for DPGME and ion mobility spectrometry: the patent that designates it requires detectability by that technique, we did not measure ion mobility, and our result says nothing about whether the compound serves that purpose. The claim at issue is narrower than either. It is that a validation performed against one measurement is being carried, without a second validation, to colorimetric and spectroscopic detection of the agents themselves.

What can be specified is the measurement that would settle the question. It is a bridging run: candidate simulants and the agents they stand in for exposed on the same plate, in the same optical channels, at the same concentrations, within one session. Such a run yields the one quantity no amount of descriptor work can supply, a response-space distance between a simulant and its agent measured under identical conditions, and it needs no new chemistry and no new instrument. It needs only that the simulant be placed beside the agent rather than in place of it. Until that run exists, a statement that a compound is a validated detection simulant reports a convention, and a convention should be declared as one.

Experimental Section

Array measurements

The response data are the colorimetric tables of three published runs of one array, all three from this laboratory and sharing authors with the present paper, reanalysed here and not remeasured. We state that here rather than only in the acknowledgement because it bears on how the reanalysis should be read: the accuracy figures this paper withdraws, and the agent production claim it corrects, are our own. The first covers tabun, sarin, soman, cyclosarin, VX and dimethyl methylphosphonate;[18] the second hydrogen cyanide, cyanogen chloride, phosgene, chloropicrin, sulfur mustard, nitrogen mustard and lewisite;[19] the third the Novichok agents A-230, A-232, A-234 and A-242.[20] We made no measurement of our own and handled no agent; what follows is what the three sources state. Each reports that the work was carried out at an Organisation for the Prohibition of Chemical Weapons designated laboratory under Chemical Weapons Convention procedures. The first names the CBRN Defense Research Institute, Seoul, as the source of the agents;[18] the third states that waste was neutralised by validated alkaline hydrolysis before disposal and does not say where its agents came from.[20] We do not describe the agents as produced at that institute, because none of the three says so. For the second run, purity was confirmed by GC–MS and exceeded 95% for every compound except lewisite, at 67%.[19] The first and third runs report no purity value for any compound, which we confirmed by reading the methods of both; the reader should treat the purity of those compounds as unstated rather than as high. Dropping lewisite changes no count reported here, on either panel and in either optical channel of the second run (Supporting Information Section S9).

The array is the same 29 commercial reagents in the same order in all three runs, commercially sourced and used as received. Supporting Information Table S3 lists them. Five are not chromogenic dyes: anthracene and pyrene are fluorophores, reduced L-glutathione is a thiol reducing agent, and 2,4-dinitrophenylhydrazine and 2-hydrazinobenzothiazole are

carbonyl derivatising reagents, so part of the array responds by covalent derivatisation and part by solvatochromic shift. The sources agree on the identities and the order; they disagree on one stock solvent, giving dimethyl sulfoxide for dye 15 in the first two runs and water in the third, which we report as published rather than resolve. Stock solutions at 100 mM were prepared in ethanol, methanol, dimethyl sulfoxide, water, tetrahydrofuran or toluene and sonicated to dissolution, then diluted into acetonitrile to a working concentration of 100 μM, a thousandfold dilution that puts the residual stock solvent at 0.1% (v/v) and places every dye in one solvent environment. Each analyte was delivered in acetonitrile by serial dilution to final concentrations of 0, 10, 50, 100 and 500 μM in a 2.0 mL dye solution. Samples were mixed without incubation and imaged at once, under visible light and under ultraviolet excitation at 365 nm, with a smartphone camera fixed in an enclosed imaging box at 22 $\pm$ 3 °C and 35 $\pm$ 5% relative humidity. Regions of interest were central 20 $\times$ 10 pixel zones; RGB values were read at five positions within each region, averaged, converted to CIE $L^*a^*b^*$, and expressed as the CIEDE2000 colour difference ΔE_{00} against the 0 μM image of the same dye in the same channel. The self-referencing removes illumination and device offsets common to an image pair. The sources do not state the working colour space, the reference white or whether the camera's transfer function was linearised before conversion, and we were not able to establish them; the deposited tables begin at ΔE_{00}, so this work inherits those three choices rather than making them. A signed readout, ΔL^*, Δa^* and Δb^* in place of a single unsigned magnitude, would also carry direction that ΔE_{00} discards, and is part of what the bridging run proposed in the Conclusions should record.

The three runs are different compounds measured on different plates and are not replicates of one another, so no ΔE_{00} value is compared across runs; every analysis below is run within a run and channel, and results are combined only at the level of correlation coefficients. The two channels are physically different measurements rather than repeats and are treated as independent modalities. The tables hold one value per compound, concentration, dye and channel.

Response patterns

Each compound-concentration cell gives a 29-dye ΔE_{00} vector. The processing is three steps and the order matters: each vector is scaled to unit length, the compound-axis mean within that concentration is subtracted, and the result is scaled to unit length again. Agreement is then the cosine. The second scaling is what breaks the zero-sum property discussed in Supporting Information Section S13. Dye columns bit-identical across every row were counted once, which leaves 23 effective channels in the ultraviolet pivot of the second run and 25 in that of the third.

Descriptors

Structural and physicochemical descriptors, 51 after excluding run metadata, were computed with RDKit from canonical SMILES. Ten are read from a three-dimensional structure and are taken as Boltzmann-weighted means over an ETKDGv3 conformer ensemble at 298 K with MMFF94 energies, or UFF for lewisite 1, the one compound without MMFF parameters; Supporting Information Section S4 reports what the two treatments do. Electronic descriptors were computed with GFN2-xTB in the gas phase, Boltzmann averaged over up to five conformers at 298.15 K, and the electronic space carries E_{HOMO}, E_{LUMO} and the dipole moment. The Parr electrophilicity index $\omega = \mu^2/2\eta$ and the condensed Fukui index at phosphorus are reported in Supporting Information Section S21 and admitted to no distance, ω because it is an exact function of the two orbital energies that accompany it. Dipole moments were recomputed at B3LYP/def2-SVPD, together with the two frontier orbital energies. The merged space, which carries the result reported above, is unchanged by that substitution. The three-coordinate electronic space is not, and we report what it does in Section rather than leaving the check described as a confirmation (Supporting Information Section S10). Identifiers and formulae were resolved against PubChem, and no property value was taken from a prediction service.

Distances and tests

Within each space the filter runs in three stages, in this order: constant columns are removed, then one of each pair correlated at $|r| > 0.95$, then any column that remains an exact algebraic function of others still present. The third stage overrides the protected set where the determining columns survive (Supporting Information Section S6). Remaining descriptors were standardised and distances taken as Euclidean, normalised by the median nearest-neighbour separation of the same space, which makes a normalised distance comparable within a panel and not between panels.

A paired comparison against the nearest interferent is confounded by group size, so we enumerated every way of designating three of the non-agents as simulants and report the rank of the actual designation as an exact p-value: $\binom{24}{3} = 2024$ designations on the primary panel and $\binom{17}{3} = 680$ on the twenty-eight-compound panel. The hypothesis is directional, so the reported p is one-sided. We give the two-sided values here rather than only the convention, because on the twenty-eight-compound panel they decide the answer: the structural space moves from 0.0279 one-sided to 0.0559 at twice the smaller tail and 0.0441 scored as distance from the null mean, and the merged space from 0.0250 to 0.0500 and 0.0279. Under the first of the two-sided conventions neither space reaches 0.05, and the one correction scheme that returns a value below 0.05 does so only on the one-sided reading. The Mantel tests of Section are two-sided throughout, since the direction of a matrix correlation is not fixed in advance by the hypothesis. A valid match requires a normalised distance of 2 or less together with a distance smaller than that to every compound outside the designated set. Sensitivity to descriptor choice was assessed by resampling the merged-space descriptors with replacement 1000 times under a fixed seed, restandardising and recomputing within each replicate; the intervals so obtained are percentile ranges over descriptor choice and not confidence intervals in the sampling sense.

Agreement between descriptor distance and response distance was tested by Mantel correlation with exact enumeration of relabellings in each run-channel cell, with the independent

runs combined by Fisher's method and the family of tests corrected by the Benjamini–Hochberg procedure. Six distance statistics and six multiplicity schemes were carried through in parallel so that the conclusion can be read against the analyst choices rather than from one of them.

Software

NumPy 2.5.3,[21] SciPy 1.18.1,[22] pandas 3.0.5,[23] scikit-learn 1.9.1,[24] Matplotlib 3.11.2,[25] RDKit 2026.03.6[26] and PySCF 2.14.0.[27] Conformer embedding used ETKDGv3,[28,29] the semiempirical electronic descriptors GFN2-xTB,[30,31] whose binary version was not recorded when the cache was built and is therefore not pinned, and the physicochemical descriptors the atomic-contribution and fragment methods implemented in RDKit.[32,33] All analyses are deterministic given the seed. The designation tests on both panels and the Mantel tests are exhaustive; the eight-simulant designation null and the random-subspace controls are sampled, at 20000 and 2000 draws from the recorded seed.

Data Availability

Analysis code, descriptor tables, the parsed regulatory values with their sources and every figure script are deposited in a public repository.[34] Every number we computed can be regenerated from it. Three classes are quoted rather than recomputed and are marked as such where they appear: values from the three source studies, values from regulatory registers, and the two cross-validated accuracies the deposited tables do not reproduce. The colorimetric input is the deposited ΔE_{00} table, published with the source studies; the conversion from plate images to those colour differences was performed there and is outside this deposit.

Acknowledgement

The colorimetric response data reanalysed here were measured in three earlier studies from this laboratory,[18–20] and we thank the staff of the CBRN Defense Research Institute who performed those measurements under Chemical Weapons Convention procedures. K.K. carried out the conceptualisation, software, formal analysis and visualisation and wrote the original draft. J.K. and J.Y. contributed to the analysis and validation, J.Y. to the original draft, M.K. to the validation and to reviewing and editing, D.-H.L. to validation, resources and supervision, and Y.J.J. to validation, supervision and project administration. This research received no specific grant from any funding agency in the public, commercial or not-for-profit sectors. The authors declare no competing financial interest.

Supporting Information Available

The Supporting Information is available free of charge at `https://pubs.acs.org/doi/XX.XXXX/acssensors.XXXXXXX`.

- Acute inhalation toxicity of the panel compounds and the gaps in it; predictive interpolation of that axis and why a model cannot fill the gaps; the decoupling test between fidelity and acute toxicity (Sections S3 and S23); conformer dependence of the shape descriptors and the reproduction gate; the density-functional check on the electrophilicity axis and on the dipole; the full permutation family under all six multiplicity schemes; dimensionality, subspace and descriptor-block weighting controls, each with its null; the measured colorimetric array response and what it does and does not establish; phosphorus-local reactivity for the six phosphorus-bearing compounds; and the structural filter (Section S27) (PDF)

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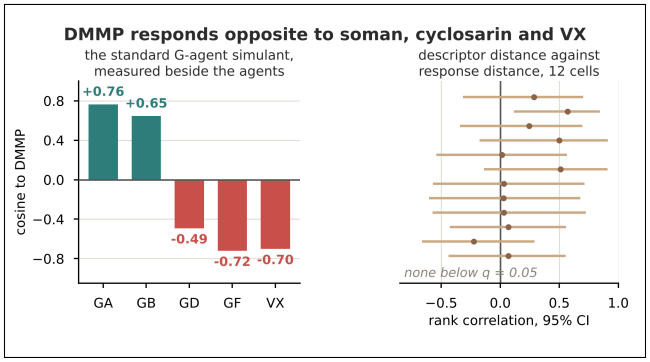
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TOC Graphic


DMMP responds opposite to soman, cyclosarin and VX
the standard G-agent simulant, measured beside the agents
descriptor distance against response distance, 12 cells
cosine to DMMP
0.8
0.4
0.0
−0.4
−0.8
+0.76
+0.65
-0.49
-0.72
-0.70
GA
GB
GD
GF
VX
none below q = 0.05
−0.5
0.0
0.5
1.0
rank correlation, 95% CI

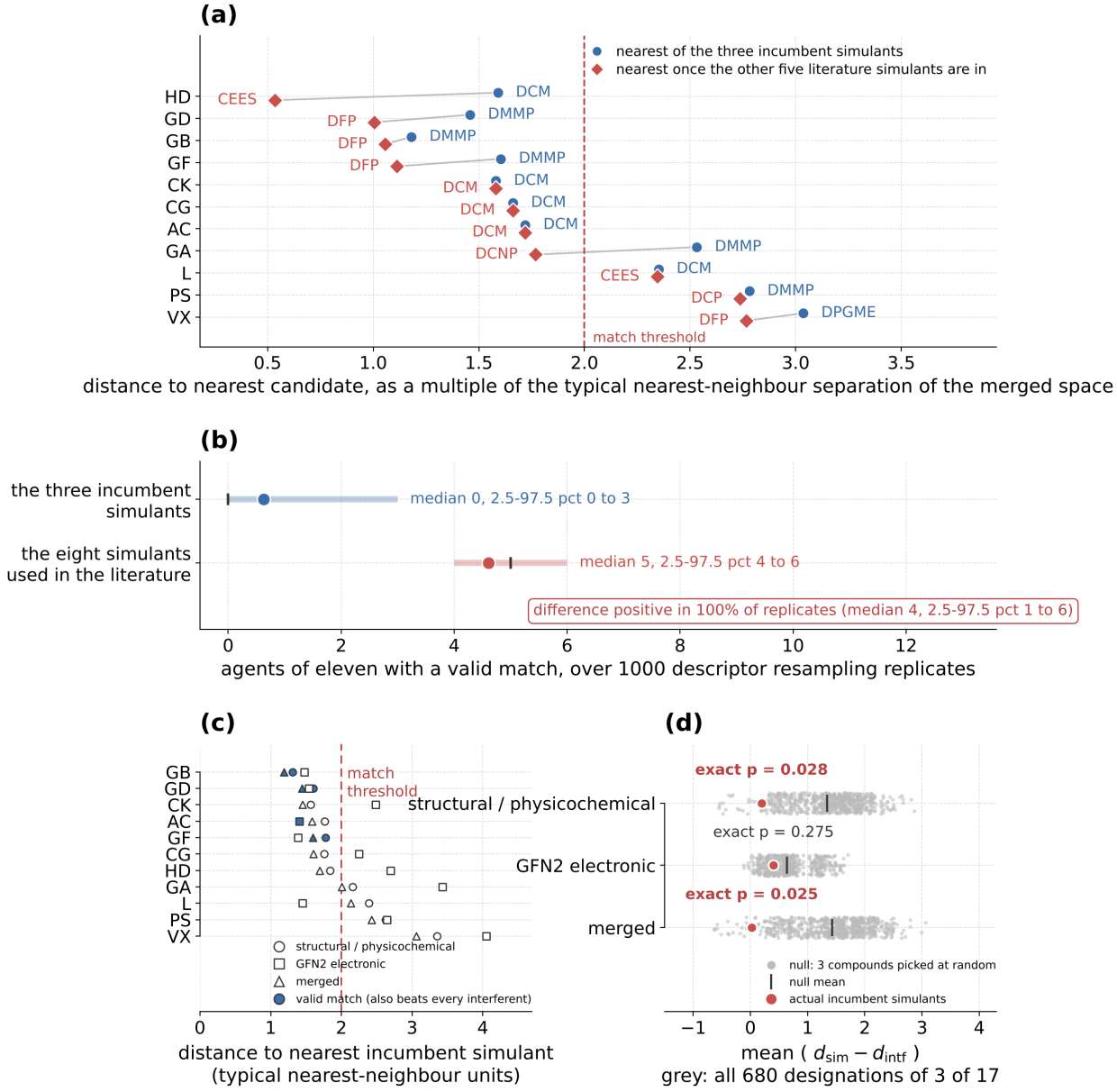


Figure 2: Descriptor-space tests. (a) Distance from each agent to its nearest candidate, for the three incumbents and for the eight compounds the sensing literature also calls simulants. (b) Valid matches over 1000 descriptor resampling replicates. (c) Distance to the nearest incumbent by space. (d) Observed mean difference against the exact null over all 680 designations of three compounds from seventeen.

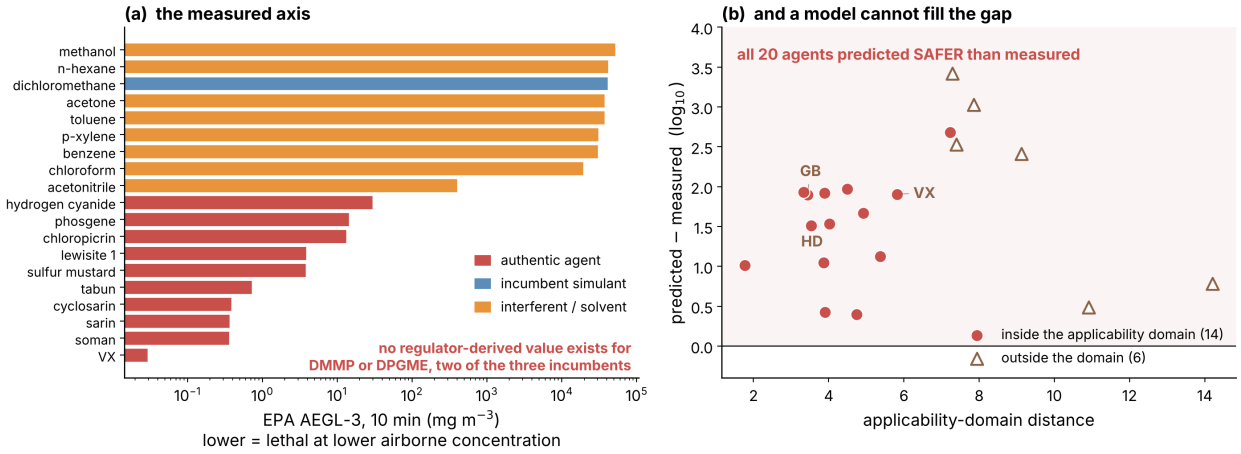


Figure 3: The toxicity axis. (a) Measured EPA AEGL-3 10-minute values for the panel compounds that carry one. (b) Prediction error of a random forest trained on 230 AEGL chemicals, for twenty military agents held out of training, against distance to the training set.